

D867

Organic Chemistry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A solvent standard bottled in amber glass has the condensed formula $\text{CH}_3\text{-CH}_2\text{-CH(CH}_3\text{)-CH}_2\text{-CH}_2\text{-CH}_3$. The supplier's label lists a purity of 99.5%. What is its IUPAC name?

- A. 3-methylhexane
- B. Heptane
- C. 2-ethylpentane
- D. 4-methylhexane

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A clear, colorless liquid stored in a flammables cabinet gives an IR spectrum with a strong, sharp peak at 1715 cm^{-1} . There is no broad band anywhere from 2500 to 3550 cm^{-1} and no small peaks near 2720 cm^{-1} . The spectrometer was serviced last month. Which functional group best fits?

- A. Alkene
- B. Ketone
- C. Carboxylic acid
- D. Alcohol

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Ethyl acetate, $\text{CH}_3\text{COOCH}_2\text{CH}_3$, a common nail-polish-remover ingredient, is dissolved in deuterated chloroform and run on a 400 MHz instrument. How many signals appear in its ^1H NMR spectrum?

- A. 8
- B. 2
- C. 4
- D. 3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A compound with formula C_2H_6O shows exactly one signal in its 1H NMR spectrum. Which structure is it?

- A. Dimethyl ether, CH_3OCH_3
- B. Methanol, CH_3OH
- C. Ethanol, CH_3CH_2OH
- D. Either one, since they share the formula C_2H_6O .

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

Bromoethane, CH_3CH_2Br , a dense, volatile liquid used as an alkylating agent, is analyzed by 1H NMR. What splitting pattern does the CH_2 signal show?

- A. A triplet, because there are three neighbors
- B. A quartet
- C. A doublet, because the CH_2 has two hydrogens
- D. A singlet, because the heavy bromine atom blocks coupling

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A 1H NMR signal appears as a triplet. How many hydrogens are on the neighboring carbons?

- A. 4
- B. 0
- C. 3
- D. 2

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A mass spectrum run on the department's older quadrupole instrument shows molecular-ion peaks at m/z 122 and m/z 124 of nearly equal height. The base peak appears at m/z 43. What does the pair of molecular-ion peaks indicate?

- A. The molecule contains one chlorine atom.
- B. The molar mass is 124 g/mol.
- C. The sample holds two different compounds.
- D. The molecule contains one bromine atom.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

tert-Butyl bromide is stirred at room temperature for two hours in a 100 mL round-bottom flask with a water-ethanol mixture. The bromide is replaced by an OH group. By which mechanism does the substitution occur?

- A. SN2, because bromide is a good leaving group
- B. SN2, because alkyl halides always react by SN2
- C. SN1, through a stable tertiary carbocation
- D. E2, because a strong base is present

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

Which substrate reacts fastest with hydroxide ion by an SN2 mechanism?

- A. C₆H₅Br (bromobenzene)
- B. (CH₃)₃CBr (tert-butyl bromide)
- C. CH₃Br (bromomethane)
- D. (CH₃)₂CHBr (2-bromopropane)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

In a demonstration planned for a unit on everyday chemistry, acetic acid and ethanol are warmed with a few drops of concentrated sulfuric acid. What are the products?

- A. Ethyl acetate and water
- B. Sodium acetate and water
- C. Diethyl ether and water
- D. No reaction; acids and alcohols do not combine.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Methyl butanoate is made by esterification. Which pair of starting materials was used?

- A. Methanoic acid and butanol
- B. Butanoic acid and methane
- C. Butanoic acid and methanol
- D. Butanol and methanol

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A teacher candidate builds butane with a class-set modeling kit purchased last spring, using black balls for carbon and white for hydrogen. She twists the model around the C2–C3 bond from the anti arrangement to the gauche arrangement. How should the two shapes be described?

- A.** Structural isomers
- B.** The same compound in two different conformations
- C.** Different compounds, since any change in shape makes a new molecule
- D.** A pair of enantiomers

Target word	The one fact that decides	Answer	☐ sure ☐ unsure ☐ guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A solvent standard ~~bottled in amber glass~~ has the condensed formula $\text{CH}_3\text{-CH}_2\text{-CH}(\text{CH}_3)\text{-CH}_2\text{-CH}_2\text{-CH}_3$. ~~The supplier's label lists a purity of 99.5%.~~ What is its IUPAC name?

Target word: IUPAC name **Decides:** Longest chain is six carbons; the methyl is on C3 when numbered from the nearer end.

A	Correct	Hexane parent; numbering from the left puts the methyl at 3, lower than 4.
B	Confusion set	Counts all seven carbons as a straight chain; that is a different (unbranched) isomer.
C	Dropped element	Uses a five-carbon chain through the branch and misses the longer six-carbon chain.
D	Inverted operation	Numbers from the far end, giving a higher locant.

2. Answer B Concept 2

A clear, colorless liquid ~~stored in a flammables cabinet~~ gives an IR spectrum with a strong, sharp peak at 1715 cm^{-1} . There is no broad band anywhere from 2500 to 3550 cm^{-1} and no small peaks near 2720 cm^{-1} . ~~The spectrometer was serviced last month.~~ Which functional group best fits?

Target word: functional group from IR **Decides:** C=O present; no O-H and no aldehyde C-H .

A	Adjacent category	C=C stretches are weaker and near 1650 cm^{-1} , not a strong 1715 cm^{-1} peak.
B	Correct	A carbonyl with no O-H band and no aldehyde C-H peaks points to a ketone.
C	Dropped element	Acids also show a very broad O-H band from $2500\text{--}3300\text{ cm}^{-1}$, which is absent.
D	Confusion set	Alcohols show a broad O-H band and no C=O peak.

3. Answer D Concept 3

Ethyl acetate, $\text{CH}_3\text{COOCH}_2\text{CH}_3$, ~~a common nail polish remover ingredient~~, is dissolved in deuterated chloroform and run on a ~~400-MHz instrument~~. How many signals appear in its ^1H NMR spectrum?

Target word: number of ^1H environments **Decides:** The acetyl CH_3 , the OCH_2 , and the ethyl CH_3 are three distinct environments.

A	Confusion set	Counts every hydrogen rather than environments.
B	Dropped element	Treats the two CH_3 groups as equivalent, but they sit in different surroundings.
C	Adjacent category	Counts carbons, which is a ^{13}C question.
D	Correct	Three sets of equivalent hydrogens give three signals.

4. Answer A Concept 3 • transfer

A compound with formula $\text{C}_2\text{H}_6\text{O}$ shows exactly one signal in its ^1H NMR spectrum. Which structure is it?

Target word: structure from signal count **Decides:** One signal means all six hydrogens are equivalent.

A	Correct	Both CH_3 groups are identical, so one signal.
B	Wrong stem number	Methanol is CH_4O , not $\text{C}_2\text{H}_6\text{O}$.
C	Confusion set	Same formula, but CH_3 , CH_2 , and OH give three signals.
D	Absolute language	Isomers with the same formula can have very different spectra.

5. Answer B Concept 4

Bromoethane, $\text{CH}_3\text{CH}_2\text{Br}$, a dense, volatile liquid used as an alkylating agent, is analyzed by ^1H NMR. What splitting pattern does the CH_2 signal show?

Target word: multiplicity of the CH_2 signal **Decides:** The CH_2 has three neighboring hydrogens (the CH_3).

A	Dropped element	Uses n instead of $n + 1$.
B	Correct	$n = 3$ neighbors, so $3 + 1 = 4$ peaks.
C	Confusion set	Counts the carbon's own hydrogens instead of its neighbors'.
D	Adjacent category	Bromine does not stop H-H coupling between adjacent carbons.

6. Answer D Concept 4 • transfer

A ^1H NMR signal appears as a triplet. How many hydrogens are on the neighboring carbons?

Target word: solve for n from the peak count **Decides:** Peaks = $n + 1$, so $n = \text{peaks} - 1$.

A	Wrong stem number	Adds 1 instead of subtracting it.
B	Absolute language	Zero neighbors gives a singlet, never a triplet.
C	Inverted operation	Treats the peak count as the neighbor count.
D	Correct	$3 \text{ peaks} - 1 = 2$ neighboring hydrogens.

7. Answer D Concept 5

A mass spectrum run on the department's older quadrupole instrument shows molecular-ion peaks at m/z 122 and m/z 124 of nearly equal height. The base peak appears at m/z 43. What does the pair of molecular-ion peaks indicate?

Target word: meaning of the $M / M+2$ pattern **Decides:** Equal M and $M+2$ heights match bromine's $^{79}\text{Br}/^{81}\text{Br}$ ratio.

A	Confusion set	Chlorine gives an $M:M+2$ ratio of about 3:1.
B	Wrong stem number	124 is the $M+2$ isotope peak, not the only molecular mass.
C	Adjacent category	An isotope pattern from one molecule, not a mixture.
D	Correct	Bromine's two isotopes are nearly 1:1, giving equal peaks two units apart.

8. Answer C Concept 6

tert-Butyl bromide is stirred at room temperature for two hours in a 100 mL round bottom flask with a water-ethanol mixture. The bromide is replaced by an OH group. By which mechanism does the substitution occur?

Target word: substitution mechanism **Decides:** Tertiary substrate in a polar protic solvent with a weak nucleophile.

A	Dropped element	Leaving group matters for both; the substrate decides here.
B	Absolute language	Tertiary halides essentially never react by $\text{S}_\text{N}2$.
C	Correct	A crowded tertiary carbon blocks backside attack but forms a stable carbocation.
D	Adjacent category	Water and ethanol are weak bases; no strong base is present.

9. Answer C Concept 6 • transfer

Which substrate reacts fastest with hydroxide ion by an SN2 mechanism?

Target word: best SN2 substrate **Decides:** SN2 rate falls as the carbon becomes more crowded.

A	Confusion set	Aryl halides do not undergo SN2 at the ring carbon.
B	Inverted operation	Most hindered; this is the best SN1 substrate, not SN2.
C	Correct	The least hindered carbon gives the easiest backside attack.
D	Adjacent category	Secondary substrates react by SN2 more slowly than methyl.

10. Answer A Concept 7

~~In a demonstration planned for a unit on everyday chemistry,~~ acetic acid and ethanol are warmed with a few drops of concentrated sulfuric acid. What are the products?

Target word: products of esterification **Decides:** Carboxylic acid + alcohol with acid catalyst.

A	Correct	Fischer esterification gives an ester and water.
B	Confusion set	That is acid-base neutralization with sodium hydroxide.
C	Adjacent category	That comes from dehydrating two ethanol molecules, not from an acid and an alcohol.
D	Absolute language	They combine readily under acid catalysis.

11. Answer C Concept 7 • transfer

Methyl butanoate is made by esterification. Which pair of starting materials was used?

Target word: work backward from ester to acid and alcohol **Decides:** “-oate” names the acid; the alkyl word names the alcohol.

A	Inverted operation	Swaps the acid and alcohol roles.
B	Confusion set	Methane is an alkane; the methyl group comes from methanol.
C	Correct	Butanoate comes from butanoic acid; methyl comes from methanol.
D	Dropped element	An ester needs a carboxylic acid; two alcohols give an ether.

12. Answer B Concept 8

~~A teacher candidate builds butane with a class-set modeling kit purchased last spring, using black balls for carbon and white for hydrogen.~~ She twists the model around the C2–C3 bond from the anti arrangement to the gauche arrangement. How should the two shapes be described?

Target word: relationship between two shapes **Decides:** Only rotation about a single bond occurred; no bonds broke.

A	Confusion set	Structural isomers differ in bonding order, which did not change.
B	Correct	Single-bond rotation interconverts conformers of one molecule.
C	Absolute language	Shape changes from rotation do not change identity.
D	Adjacent category	Enantiomers are non-superimposable mirror images that need bond breaking to interconvert.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	10	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Parent chain and lowest locants	Name from the longest continuous carbon chain, numbered from the end that gives substituents the lowest numbers.	Taking the chain as drawn in a straight line, or numbering from the wrong end.	A branched condensed formula to be named.
2	Reading an IR spectrum	A strong peak near 1700–1750 cm^{-1} means C=O; a broad band near 3200–3550 cm^{-1} means O–H; the absence of a peak rules a group out.	Naming a carboxylic acid or alcohol from the carbonyl peak alone.	Wavenumbers listed with “present” and “absent” bands.
3	Counting ^1H NMR signals	Each set of chemically equivalent hydrogens gives one signal; the count is of environments, not atoms.	Counting every hydrogen or every carbon.	“How many signals?” or a formula with one signal reported.
4	Splitting: the n + 1 rule	A signal is split into n + 1 peaks, where n is the number of hydrogens on neighboring carbons.	Counting the hydrogens on the carbon itself, or using n instead of n + 1.	Doublet, triplet, quartet in the stem.
5	Mass spectrum isotope patterns	M and M+2 peaks of equal height signal one bromine; a 3:1 ratio signals one chlorine.	Reading the M+2 peak as the molar mass or as a second compound.	Two molecular-ion peaks two units apart.
6	SN1 vs. SN2	SN2 needs an open back side (methyl > primary); tertiary substrates in polar protic solvent react by SN1 through a carbocation.	Deciding the mechanism by the leaving group alone.	Substrate is labeled primary, secondary, or tertiary.
7	Esterification	Carboxylic acid + alcohol (acid catalyst) $\rightarrow$ ester + water; the acid supplies the “-oate” part and the alcohol supplies the alkyl part.	Swapping which piece came from the acid and which from the alcohol.	An acid, an alcohol, and a few drops of H_2SO_4 .
8	Conformations vs. isomers	Rotating around a single bond changes conformation, not identity; isomers require breaking bonds to interconvert.	Calling two model shapes different compounds because they look different.	A modeling kit twisted about a C–C single bond.

Course scope (public description): This Organic Chemistry course prepares teacher candidates to understand and apply foundational concepts in carbon-based chemistry. You will learn to describe molecular structures using proper nomenclature, analyze chemical processes, and apply instrumental methods of analysis to determine molecular structures. The course emphasizes practical skills, including the use of modeling kits and spectroscopic techniques to synthesize and analyze organic compounds. By the end of this course, you will be equipped to teach these concepts effectively to your students.